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Lab Assignment-7

Questions

1. Create a view named patient_basic_info showing patient name, gender, and city.

```
mysql> create view patient_basic_info AS
    -> Select patient_name, gender , city from Patients;
Query OK, 0 rows affected (0.02 sec)

mysql> select * from patient_basic_info;
+------------------+--------+---------+
| patient_name     | gender | city    |
+------------------+--------+---------+
| Amit Sharma      | Male   | Delhi   |
| Neha Singh       | Female | Mumbai  |
| Rahul Das        | Male   | Kolkata |
| Priya Verma      | Female | Chennai |
| Karan Mehta      | Male   | Delhi   |
| Ratnesh Srivastwa | Male   | Patna   |
+------------------+--------+---------+
6 rows in set (0.01 sec)
```

2. Create a view named doctor_fee_view showing doctor name and consultation fee.

```
mysql> create view doctor_fee_view AS
    -> select  doctor_name,consultation_fee from Doctors;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from doctor_fee_view;
+-------------+------------------+
| doctor_name | consultation_fee |
+-------------+------------------+
| Dr. Kumar   |              800 |
| Dr. Meena   |              500 |
| Dr. Shah    |             1000 |
| Dr. Roy     |              700 |
+-------------+------------------+
4 rows in set (0.00 sec)
```

3. Create a view showing all completed appointments.

```
mysql> create view completed_appointments AS
    -> select appointment_id, patient_id, doctor_id, appointment_date, status from Appointments where status = "Completed";
Query OK, 0 rows affected (0.02 sec)

mysql> select * from completed_appointments;
+----------------+------------+-----------+------------------+-----------+
| appointment_id | patient_id | doctor_id | appointment_date | status    |
+----------------+------------+-----------+------------------+-----------+
|           1001 |          1 |       101 | 2026-02-01       | Completed |
|           1003 |          3 |       101 | 2026-02-10       | Completed |
|           1005 |          5 |       104 | 2026-02-20       | Completed |
+----------------+------------+-----------+------------------+-----------+
3 rows in set (0.00 sec)
```

4. Create a view named appointment_details showing patient name, doctor name, appointment date, and status.

```
mysql> select p.patient_name, d.doctor_name, a.appointment_date, a.status from appointments a JOIN Patients p ON a.patient_id = p.patient_id JOIN Doctors d
ON a.patient_id = p.patient_id;
+-------------+-------------+-----------------+----------+
| patient_name | doctor_name | appointment_date | status    |
+-------------+-------------+-----------------+----------+
| Amit Sharma  | Dr. Roy     | 2026-02-01       | Completed |
| Amit Sharma  | Dr. Shah    | 2026-02-01       | Completed |
| Amit Sharma  | Dr. Meena   | 2026-02-01       | Completed |
| Amit Sharma  | Dr. Kumar   | 2026-02-01       | Completed |
| Amit Sharma  | Dr. Roy     | 2026-02-25       | Pending   |
| Amit Sharma  | Dr. Shah    | 2026-02-25       | Pending   |
| Amit Sharma  | Dr. Meena   | 2026-02-25       | Pending   |
| Amit Sharma  | Dr. Kumar   | 2026-02-25       | Pending   |
| Neha Singh   | Dr. Roy     | 2026-02-05       | Pending   |
| Neha Singh   | Dr. Shah    | 2026-02-05       | Pending   |
| Neha Singh   | Dr. Meena   | 2026-02-05       | Pending   |
| Neha Singh   | Dr. Kumar   | 2026-02-05       | Pending   |
| Rahul Das    | Dr. Roy     | 2026-02-10       | Completed |
| Rahul Das    | Dr. Shah    | 2026-02-10       | Completed |
| Rahul Das    | Dr. Meena   | 2026-02-10       | Completed |
| Rahul Das    | Dr. Kumar   | 2026-02-10       | Completed |
| Priya Verma  | Dr. Roy     | 2026-02-15       | Cancelled |
| Priya Verma  | Dr. Shah    | 2026-02-15       | Cancelled |
| Priya Verma  | Dr. Meena   | 2026-02-15       | Cancelled |
| Priya Verma  | Dr. Kumar   | 2026-02-15       | Cancelled |
| Karan Mehta  | Dr. Roy     | 2026-02-20       | Completed |
| Karan Mehta  | Dr. Shah    | 2026-02-20       | Completed |
| Karan Mehta  | Dr. Meena   | 2026-02-20       | Completed |
| Karan Mehta  | Dr. Kumar   | 2026-02-20       | Completed |
+-------------+-------------+-----------------+----------+
24 rows in set (0.02 sec)
```

5. Create a view showing only appointments with Cardiologist.

```
mysql> CREATE VIEW cardiologist_appointments AS
    -> SELECT a.* FROM Appointments a
    -> JOIN Doctors d ON a.doctor_id = d.doctor_id
    -> WHERE d.specialization = 'Cardiologist';
Query OK, 0 rows affected (0.01 sec)

mysql> select * from cardiologist_appointments;
+----------------+------------+-----------+------------------+-----------+
| appointment_id | patient_id | doctor_id | appointment_date | status    |
+----------------+------------+-----------+------------------+-----------+
|           1001 |          1 |       101 | 2026-02-01       | Completed |
|           1003 |          3 |       101 | 2026-02-10       | Completed |
+----------------+------------+-----------+------------------+-----------+
2 rows in set (0.00 sec)
```

6. Create a view showing patients from Delhi.

```
mysql> create view Delhi_patients AS
    -> select  patient_id, patient_name, city from Patients where city = "Delhi";
Query OK, 0 rows affected (0.01 sec)

mysql> select * from Delhi_patients:
    -> :
    -> ^C
mysql> select * from Delhi_patients;
+------------+--------------+-------+
| patient_id | patient_name | city  |
+------------+--------------+-------+
|          1 | Amit Sharma  | Delhi |
|          5 | Karan Mehta  | Delhi |
+------------+--------------+-------+
2 rows in set (0.01 sec)

mysql> |
```

7. Create a view showing doctors whose consultation fee is greater than 700.

```
mysql> create view Fee AS
    -> select doctor_id, doctor_name, specialization, consultation_fee from Doctors where consultation_fee>700;
Query OK, 0 rows affected (0.02 sec)

mysql> select * from Fee;
+-----------+-------------+----------------+------------------+
| doctor_id | doctor_name | specialization | consultation_fee |
+-----------+-------------+----------------+------------------+
|       101 | Dr. Kumar   | Cardiologist   |              800 |
|       103 | Dr. Shah    | Neurologist    |             1000 |
+-----------+-------------+----------------+------------------+
2 rows in set (0.00 sec)

mysql> |
```

8. Create a view showing patient name, doctor name, and consultation fee.

```
mysql> CREATE VIEW patient_doctor_fee AS
    -> SELECT p.patient_name, d.doctor_name, d.consultation_fee
    -> FROM Appointments a
    -> JOIN Patients p ON a.patient_id=p.patient_id
    -> JOIN Doctors d ON a.doctor_id=d.doctor_id;
Query OK, 0 rows affected (0.03 sec)

mysql> select * from patient-doctor_fee;
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r '-doctor_fee' at line 1
mysql> select * from patient_doctor_fee;
+--------------+-------------+------------------+
| patient_name | doctor_name | consultation_fee |
+--------------+-------------+------------------+
| Amit Sharma  | Dr. Kumar   |              800 |
| Rahul Das    | Dr. Kumar   |              800 |
| Neha Singh   | Dr. Meena   |              500 |
| Priya Verma  | Dr. Shah    |             1000 |
| Amit Sharma  | Dr. Shah    |             1000 |
| Karan Mehta  | Dr. Roy     |              700 |
+--------------+-------------+------------------+
6 rows in set (0.02 sec)
```

9. Create a view showing total number of appointments per doctor.

```
mysql> CREATE VIEW total_appointments_per_doctor AS
    -> SELECT d.doctor_name, COUNT(a.appointment_id) AS total_appointments
    -> FROM Doctors d
    -> LEFT JOIN Appointments a ON d.doctor_id=a.doctor_id
    -> GROUP BY d.doctor_name;
Query OK, 0 rows affected (0.01 sec)

mysql> select * from ^C
mysql> select * from total_appointments_per_doctor;
+-------------+--------------------+
| doctor_name | total_appointments |
+-------------+--------------------+
| Dr. Kumar   |                  2 |
| Dr. Meena   |                  1 |
| Dr. Shah    |                  2 |
| Dr. Roy     |                  1 |
+-------------+--------------------+
4 rows in set (0.02 sec)
```

10. Create a view showing pending appointments only.

```
mysql> CREATE VIEW pending_appointments AS SELECT * FROM Appointments WHERE
status = 'Pending';
Query OK, 0 rows affected (0.02 sec)
```

```
mysql> select * from pending_appointments;
+----------------+------------+-----------+------------------+---------+
| appointment_id | patient_id | doctor_id | appointment_date | status  |
+----------------+------------+-----------+------------------+---------+
|           1002 |          2 |       102 | 2026-02-05       | Pending |
|           1006 |          1 |       103 | 2026-02-25       | Pending |
+----------------+------------+-----------+------------------+---------+
2 rows in set (0.00 sec)
```

11. Create a stored procedure to display all patients.

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE ShowAllPatients()
    -> BEGIN
    ->     SELECT * FROM Patients;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL ShowAllPatients()
    -> ;
+------------+------------------+------+--------+---------+
| patient_id | patient_name     | age  | gender | city    |
+------------+------------------+------+--------+---------+
|          1 | Amit Sharma      |   30 | Male   | Delhi   |
|          2 | Neha Singh       |   25 | Female | Mumbai  |
|          3 | Rahul Das        |   40 | Male   | Kolkata |
|          4 | Priya Verma      |   35 | Female | Chennai |
|          5 | Karan Mehta      |   28 | Male   | Delhi   |
|          6 | Ratnesh Srivastwa |  24 | Male   | Patna   |
+------------+------------------+------+--------+---------+
6 rows in set (0.00 sec)

Query OK, 0 rows affected (0.02 sec)
```

12. Create a stored procedure to display all doctors.

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE ShowAllDoctors()
    -> BEGIN
    ->     SELECT * FROM Doctors;
    -> END //
Query OK, 0 rows affected (0.07 sec)

mysql>
mysql> DELIMITER ;
mysql> call PROCEDURE ShowAllDoctors();
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r 'PROCEDURE ShowAllDoctors()' at line 1
mysql> call  ShowAllDoctors();
+-----------+-------------+----------------+------------------+
| doctor_id | doctor_name | specialization | consultation_fee |
+-----------+-------------+----------------+------------------+
|       101 | Dr. Kumar   | Cardiologist   |              800 |
|       102 | Dr. Meena   | Dermatologist  |              500 |
|       103 | Dr. Shah    | Neurologist    |             1000 |
|       104 | Dr. Roy     | Orthopedic     |              700 |
+-----------+-------------+----------------+------------------+
4 rows in set (0.01 sec)
```

13. Create a stored procedure to show all appointments.

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE ShowAllAppointments()
    -> BEGIN
    ->     SELECT * FROM Appointments;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL ShowAllAppointment();
ERROR 1305 (42000): PROCEDURE ratnesh_assignment7.ShowAllAppointment does no
t exist
mysql> CALL ShowAllAppointments();
+----------------+------------+-----------+------------------+-----------+
| appointment_id | patient_id | doctor_id | appointment_date | status    |
+----------------+------------+-----------+------------------+-----------+
|           1001 |          1 |       101 | 2026-02-01       | Completed |
|           1002 |          2 |       102 | 2026-02-05       | Pending   |
|           1003 |          3 |       101 | 2026-02-10       | Completed |
|           1004 |          4 |       103 | 2026-02-15       | Cancelled |
|           1005 |          5 |       104 | 2026-02-20       | Completed |
|           1006 |          1 |       103 | 2026-02-25       | Pending   |
+----------------+------------+-----------+------------------+-----------+
6 rows in set (0.01 sec)

Query OK, 0 rows affected (0.04 sec)

mysql>
```

14. Create a stored procedure that shows all appointments of a given patient_id.

```
mysql> CREATE PROCEDURE GetPatientAppointments(IN p_id INT)
    -> BEGIN
    ->     SELECT * FROM Appointments
    ->     WHERE patient_id = p_id;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetPatientAppointments(IN p_id INT);
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual t
hat corresponds to your MySQL server version for the right syntax to use nea
r 'IN p_id INT)' at line 1
mysql> CALL GetPatientAppointments(1);
+----------------+------------+-----------+------------------+-----------+
| appointment_id | patient_id | doctor_id | appointment_date | status    |
+----------------+------------+-----------+------------------+-----------+
|           1001 |          1 |       101 | 2026-02-01       | Completed |
|           1006 |          1 |       103 | 2026-02-25       | Pending   |
+----------------+------------+-----------+------------------+-----------+
2 rows in set (0.02 sec)
```

15. Create a stored procedure that shows doctors of a given specialization.

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetDoctorsBySpecialization(IN spec_name VARCHAR(50))

    -> BEGIN
    ->     SELECT * FROM Doctors
    ->     WHERE specialization = spec_name;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetDoctorsBySpecialization(cardiologist);
ERROR 1054 (42S22): Unknown column 'cardiologist' in 'field list'
mysql> CALL GetDoctorsBySpecialization( Cardiologist);
ERROR 1054 (42S22): Unknown column 'Cardiologist' in 'field list'
mysql> CALL GetDoctorsBySpecialization('Cardiologist');
+-----------+-------------+----------------+------------------+
| doctor_id | doctor_name | specialization | consultation_fee |
+-----------+-------------+----------------+------------------+
|       101 | Dr. Kumar   | Cardiologist   |              800 |
+-----------+-------------+----------------+------------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

16. Create a stored procedure to show appointments by status (input parameter).

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetAppointmentsByStatus(IN appt_status VARCHAR(20))
    -> BEGIN
    ->     SELECT * FROM Appointments
    ->     WHERE status = appt_status;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetAppointmentsByStatus('Pending');
+----------------+------------+-----------+------------------+---------+
| appointment_id | patient_id | doctor_id | appointment_date | status  |
+----------------+------------+-----------+------------------+---------+
|           1002 |          2 |       102 | 2026-02-05       | Pending |
|           1006 |          1 |       103 | 2026-02-25       | Pending |
+----------------+------------+-----------+------------------+---------+
2 rows in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)

mysql> CALL GetAppointmentsByStatus('Complete');
Empty set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)

mysql> CALL GetAppointmentsByStatus('Completed');
+----------------+------------+-----------+------------------+-----------+
| appointment_id | patient_id | doctor_id | appointment_date | status    |
+----------------+------------+-----------+------------------+-----------+
|           1001 |          1 |       101 | 2026-02-01       | Completed |
|           1003 |          3 |       101 | 2026-02-10       | Completed |
|           1005 |          5 |       104 | 2026-02-20       | Completed |
+----------------+------------+-----------+------------------+-----------+
3 rows in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

17. Create a stored procedure to count total number of patients.

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE CountTotalPatients()
    -> BEGIN
    ->     SELECT COUNT(*) AS Total_Patients FROM Patients;
    -> END //
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql> DELIMITER ;
mysql> call CountTotalPatients();
+----------------+
| Total_Patients |
+----------------+
|              6 |
+----------------+
1 row in set (0.02 sec)

Query OK, 0 rows affected (0.03 sec)
```

18. Create a stored procedure to calculate total revenue generated (sum of consultation fees for completed appointments).

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE CalculateTotalRevenue()
    -> BEGIN
    ->     SELECT SUM(d.consultation_fee) AS Total_Revenue
    ->     FROM Appointments a
    ->     JOIN Doctors d ON a.doctor_id = d.doctor_id
    ->     WHERE a.status = 'Completed';
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> call CalculateTotalRevenue();
+---------------+
| Total_Revenue |
+---------------+
|          2300 |
+---------------+
1 row in set (0.01 sec)

Query OK, 0 rows affected (0.01 sec)
```

19. Create a stored procedure to display doctor name and total number of appointments handled.

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE ShowDoctorAppointmentCounts()
    -> BEGIN
    ->     SELECT
    ->         d.doctor_name,
    ->         COUNT(a.appointment_id) AS Total_Appointments
    ->     FROM
    ->         Doctors d
    ->     LEFT JOIN
    ->         Appointments a ON d.doctor_id = a.doctor_id
    ->     GROUP BY
    ->         d.doctor_id, d.doctor_name;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> call ShowDoctorAppointmentCounts();
+-------------+--------------------+
| doctor_name | Total_Appointments |
+-------------+--------------------+
| Dr. Kumar   |                  2 |
| Dr. Meena   |                  1 |
| Dr. Shah    |                  2 |
| Dr. Roy     |                  1 |
+-------------+--------------------+
4 rows in set (0.01 sec)

Query OK, 0 rows affected (0.02 sec)
```

20. Create a stored procedure to display patient name and doctor name for a given appointment_id.

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE GetAppointmentNames(IN appt_id INT)
    -> BEGIN
    ->     SELECT
    ->         p.patient_name,
    ->         d.doctor_name
    ->     FROM
    ->         Appointments a
    ->     JOIN
    ->         Patients p ON a.patient_id = p.patient_id
    ->     JOIN
    ->         Doctors d ON a.doctor_id = d.doctor_id
    ->     WHERE
    ->         a.appointment_id = appt_id;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL GetAppointmentNames(1001);
+--------------+-------------+
| patient_name | doctor_name |
+--------------+-------------+
| Amit Sharma  | Dr. Kumar   |
+--------------+-------------+
1 row in set (0.00 sec)

Query OK, 0 rows affected (0.01 sec)
```

21. Create a procedure to insert a new patient.

```
mysql> DELIMITER //
mysql>
mysql>
mysql>
mysql> CREATE PROCEDURE AddPatient(
    ->
    ->     IN p_id INT,
    ->
    ->     IN p_name VARCHAR(50),
    ->
    ->     IN p_age INT,
    ->
    ->     IN p_gender VARCHAR(10),
    ->
    ->     IN p_city VARCHAR(50)
    ->
    -> )
    ->
    -> BEGIN
    ->
    ->     INSERT INTO Patients (patient_id, patient_name, age, gender, city)
    ->
    ->     VALUES (p_id, p_name, p_age, p_gender, p_city);
    ->
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql>
mysql>
mysql> DELIMITER ;
mysql> call AddPatient(7, 'Himanshu', 25, 'Male',  'Patna');
ERROR 1062 (23000): Duplicate entry '7' for key 'patients.PRIMARY'
mysql> call AddPatient(8, 'Himanshu Ranjan', 25, 'Male',  'Noida');
Query OK, 1 row affected (0.01 sec)

mysql> CALL ShowAllPatients();
+------------+-------------------+------+--------+---------+
| patient_id | patient_name      | age  | gender | city    |
+------------+-------------------+------+--------+---------+
|          1 | Amit Sharma       |   30 | Male   | Delhi   |
|          2 | Neha Singh        |   25 | Female | Mumbai  |
|          3 | Rahul Das         |   40 | Male   | Kolkata |
|          4 | Priya Verma       |   35 | Female | Chennai |
|          5 | Karan Mehta       |   28 | Male   | Delhi   |
|          6 | Ratnesh Srivastwa |   24 | Male   | Patna   |
|          7 | Himanshu          |   25 | Male   | Patna   |
|          8 | Himanshu Ranjan   |   25 | Male   | Noida   |
+------------+-------------------+------+--------+---------+
8 rows in set (0.00 sec)
```

22. Create a procedure to update consultation fee of a doctor.

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE UpdateDoctorFee(
    ->     IN d_id INT,
    ->     IN new_fee INT
    -> )
    -> BEGIN
    ->     UPDATE Doctors
    ->     SET consultation_fee = new_fee
    ->     WHERE doctor_id = d_id;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL UpdateDoctorFee(101, 900);
Query OK, 1 row affected (0.01 sec)

mysql> call ShowAllDoctors();
+-----------+-------------+----------------+------------------+
| doctor_id | doctor_name | specialization | consultation_fee |
+-----------+-------------+----------------+------------------+
|       101 | Dr. Kumar   | Cardiologist   |              900 |
|       102 | Dr. Meena   | Dermatologist  |              500 |
|       103 | Dr. Shah    | Neurologist    |             1000 |
|       104 | Dr. Roy     | Orthopedic     |              700 |
+-----------+-------------+----------------+------------------+
4 rows in set (0.00 sec)

Query OK, 0 rows affected (0.02 sec)
```

23. Create a procedure to delete an appointment.

```
mysql> DELIMITER //
mysql>
mysql> CREATE PROCEDURE DeleteAppointment(IN appt_id INT)
    -> BEGIN
    ->     DELETE FROM Appointments
    ->     WHERE appointment_id = appt_id;
    -> END //
Query OK, 0 rows affected (0.01 sec)

mysql>
mysql> DELIMITER ;
mysql> CALL DeleteAppointment(1004);
Query OK, 1 row affected (0.01 sec)

mysql> call ShowAllAppointments()
    -> ;
+----------------+------------+-----------+------------------+-----------+
| appointment_id | patient_id | doctor_id | appointment_date | status    |
+----------------+------------+-----------+------------------+-----------+
|           1001 |          1 |       101 | 2026-02-01       | Completed |
|           1002 |          2 |       102 | 2026-02-05       | Pending   |
|           1003 |          3 |       101 | 2026-02-10       | Completed |
|           1005 |          5 |       104 | 2026-02-20       | Completed |
|           1006 |          1 |       103 | 2026-02-25       | Pending   |
+----------------+------------+-----------+------------------+-----------+
5 rows in set (0.00 sec)

Query OK, 0 rows affected (0.02 sec)

mysql> |
```